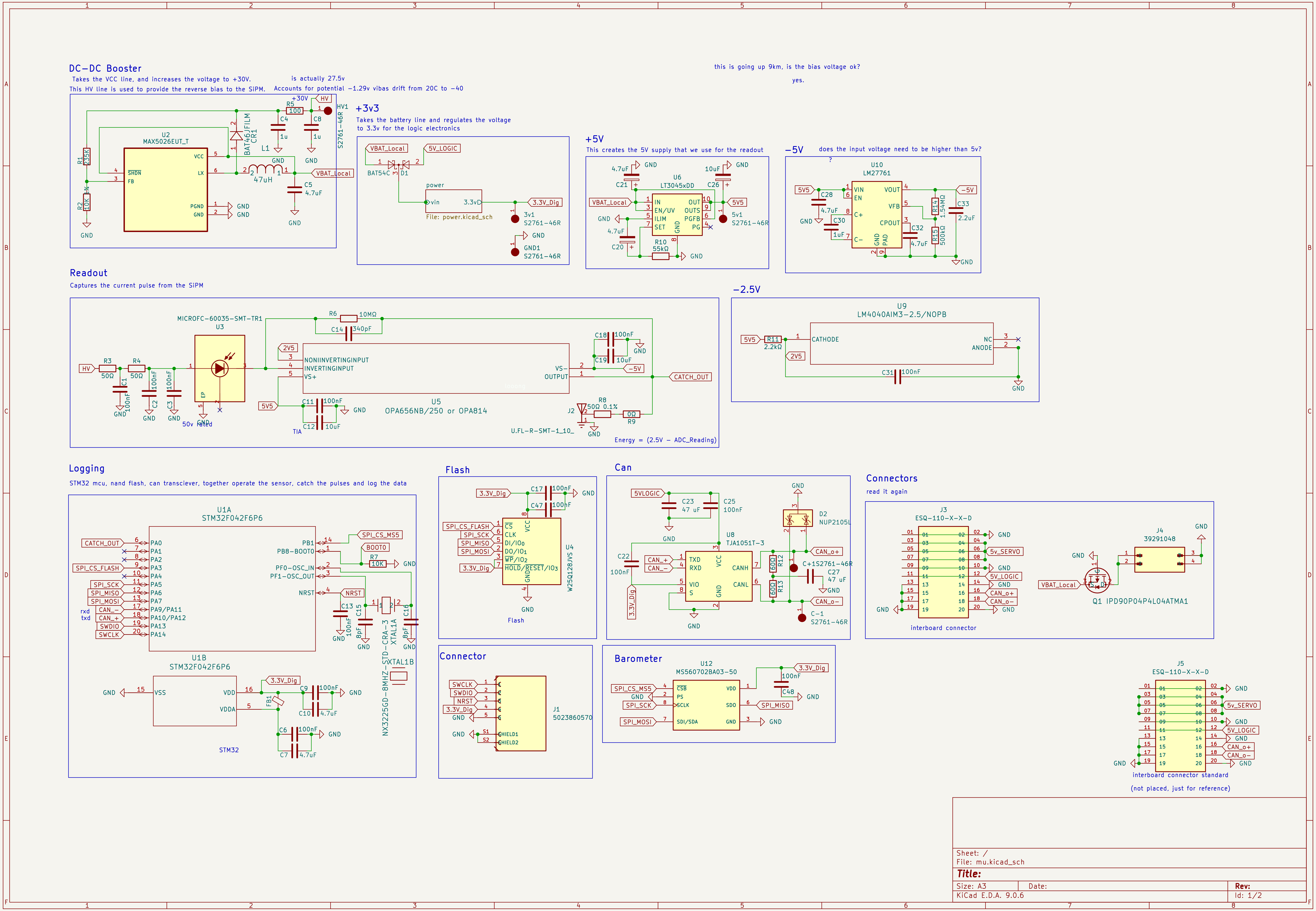
DC-DC Booster
Takes the VCC line, and increases the voltage to +30V.
This HV line is used to provide the reverse bias to the SiPM.
is actually 27.5v
Accounts for potential -1.29v vibas drift from 20C to -40
U2
MAX5026EUT_T
+3v3
Takes the battery line and regulates the voltage
to 3.3v for the logic electronics
File: power.kicad_sch
+5V
This creates the 5V supply that we use for the readout
U6
LT3045xDD
-5V
does the input voltage need to be higher than 5v?
?
U10
LM27761
this is going up 9km, is the bias voltage ok?
yes.
Readout
Captures the current pulse from the SiPM
MICROFC-60035-SMT-TR1
U3
U5
OPA656NB/250 or OPA814
TIA
50v rated
U.FL-R-SMT-1_10_
Energy = (2.5V - ADC_Reading)
-2.5V
U9
LM4040AIM3-2.5/NOPB
Logging
STM32 mcu, nand flash, can transciever, together operate the sensor, catch the pulses and log the data
U1A
STM32F042F6P6
U1B
STM32F042F6P6
STM32
NX3225GD-8MHZ-STD-CRA-3
Flash
U4
W25Q128JVS
Can
U8
TJA1051T-3
D2
NUP2105L
Connectors
read it again
J3
ESQ-110-X-X-D
interboard connector
J4
39291048
Q1 IPD90P04P4L04ATMA1
Connector
J1
5023860570
Barometer
U12
MS560702BA03-50
J5
ESQ-110-X-X-D
interboard connector standard
(not placed, just for reference)
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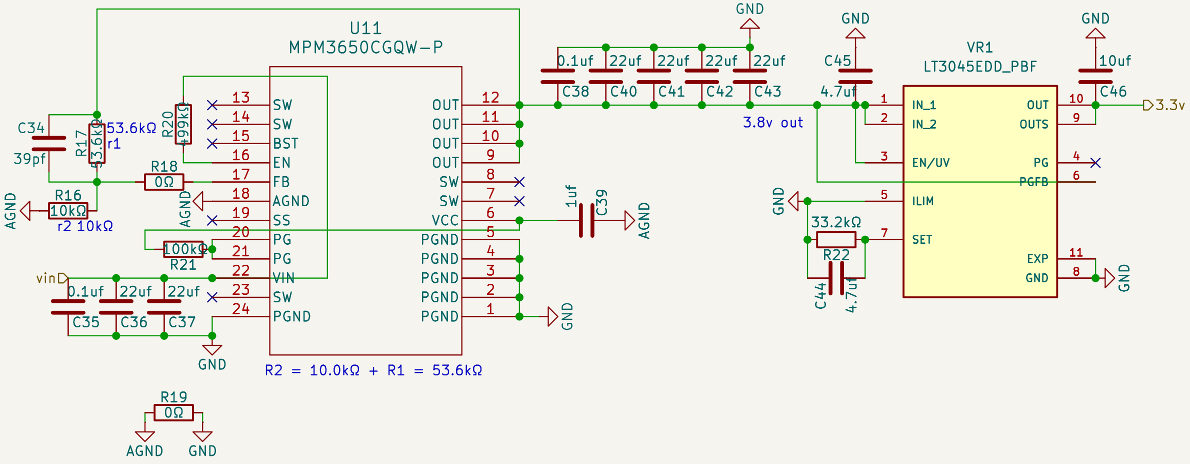
Rev:
KiCad E.D.A. 9.0.6
Id: 1/2



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